

Alien *Aedes* species: from monitoring to dissemination with a participatory approach

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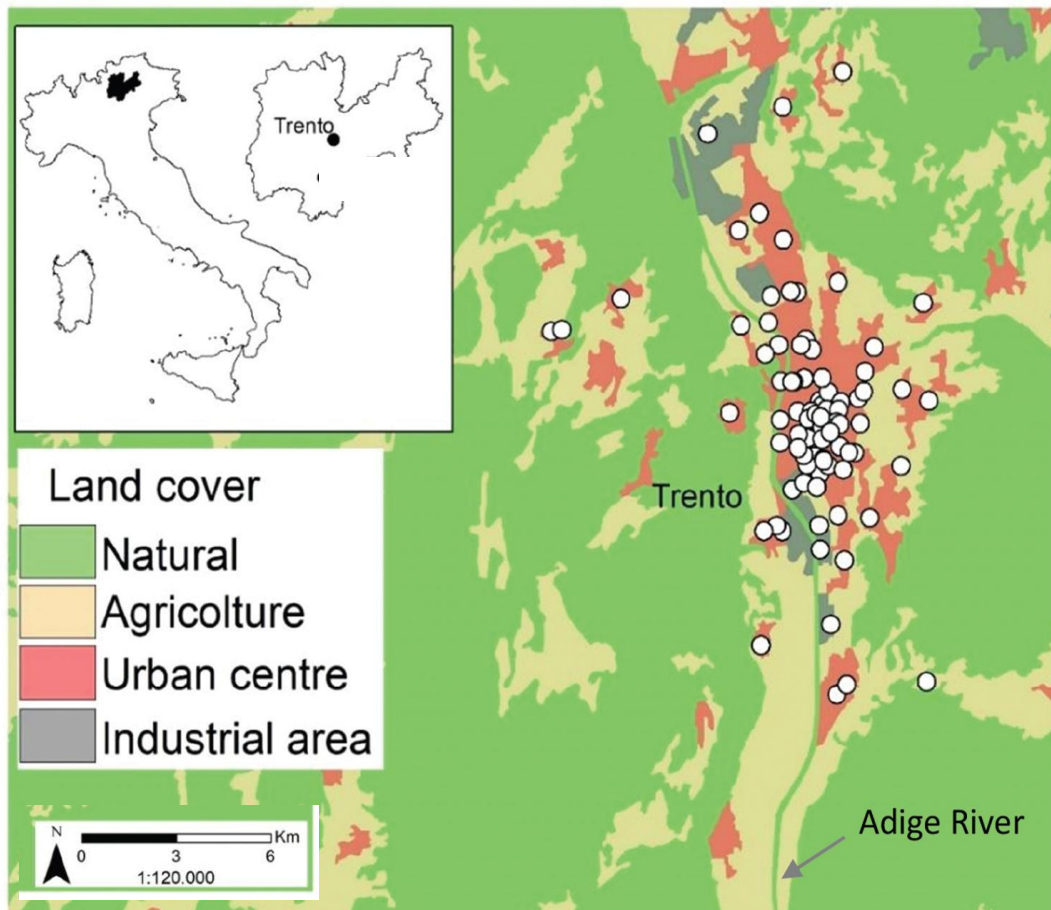


Provincia
Autonoma
di Trento



Two decades of *Aedes koreicus* in Europe:
Bridging Knowledge Gaps in Ecology and Vector Role
February 10, 2026 – Trento, Italy

A local case study: Alien *Aedes* monitoring in Trento – 16 years (2010-2025)



MUSE

Monitoring and
Scientific Research



**Municipality
of Trento**

Administration & Pest Control



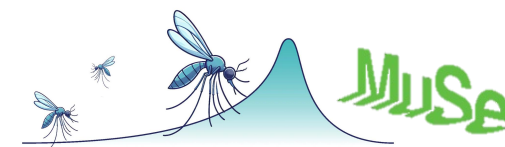
**Provincial
Health Agency**
Sanitary Control



Citizens

Mosquito Alert App
Home OVI management

Shared Goal
Knowledge
Control
Prevention



Case study: Alien *Aedes* monitoring in Trento – trapping methods

Ovitrap (since 2010)

Urbanized area <600 m asl



BG-Sentinel (since 2018) –

Gathering places (schools, hospitals)



BG-GAT (since 2021) -

Nursery schools, kindergartens



Provincial dataset
(since 2019)

Piano
di monitoraggio della
Provincia Autonoma di Trento
delle zanzare
e di altri vettori



Provincial “One Health” Committee

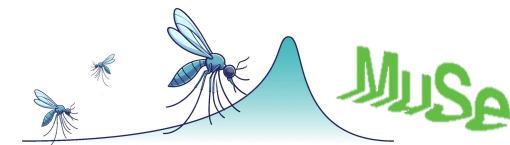


Positioning of the traps

Ovitraps were deployed according to the known **breeding site preferences** and **host presence** of *Ae. albopictus* – 7 habitat types

- ***Gathering places:** e.g. nurseries, schools, hospitals, civic centers, retirement homes
- **Residential areas**
- **Urban parks**
- **Areas near waterways**
- **Car parks**
- **Industrial areas**
- **Croplands**

BG: only in gathering places



Monitoring area based on probability classes of presence of *Ae. albopictus* >260 sites

2010-2025 - Standard monitoring (Trento Municipality <600 m asl):

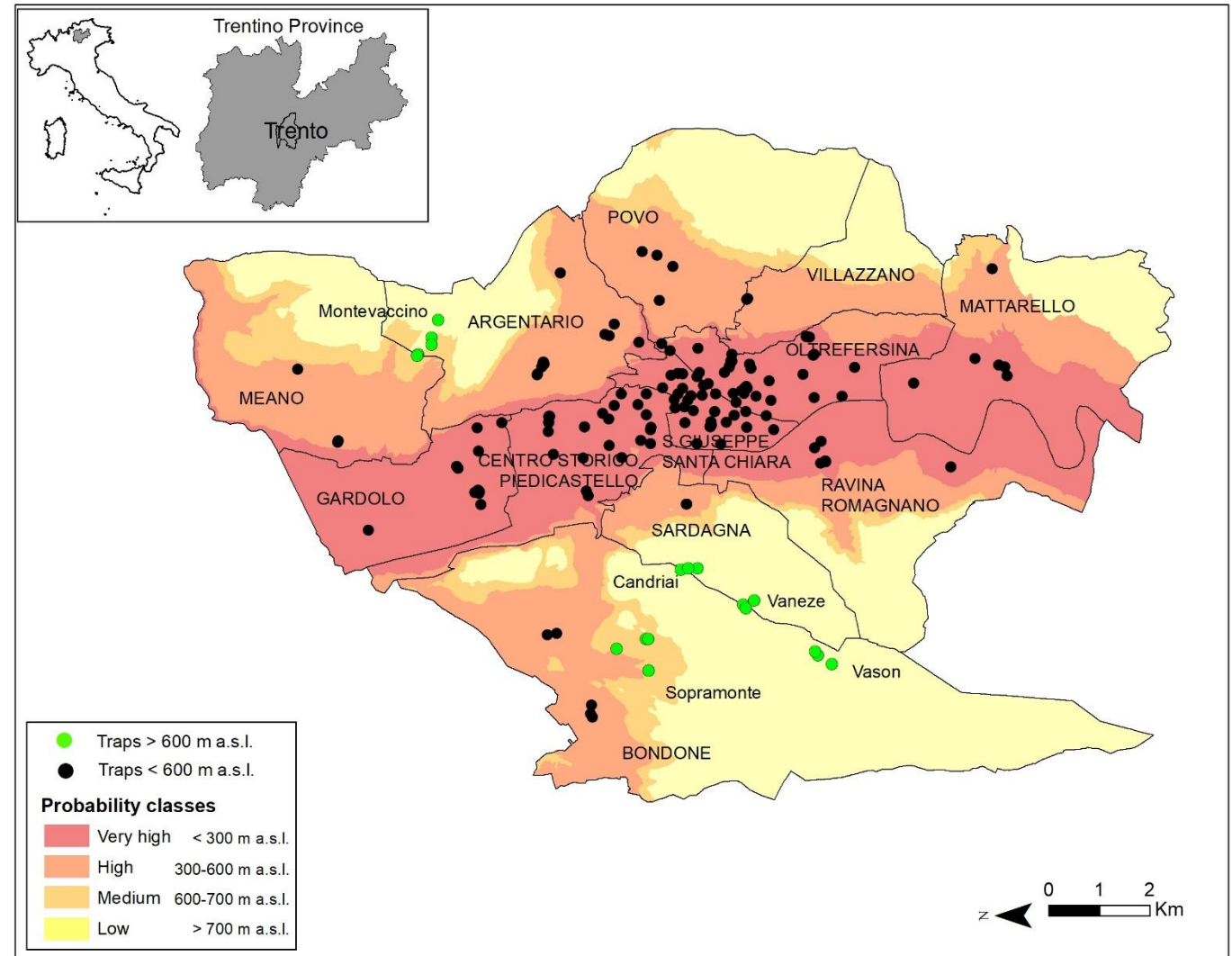
- OVI: N. 47, since 2018: 79 km² – overall 86 sites since 2010, 8 districts
- BG-Sentinel: N. 6, since 2018

2023-2025

- BG-GAT: in N. 157 in 46 nursery schools

2018-2023:

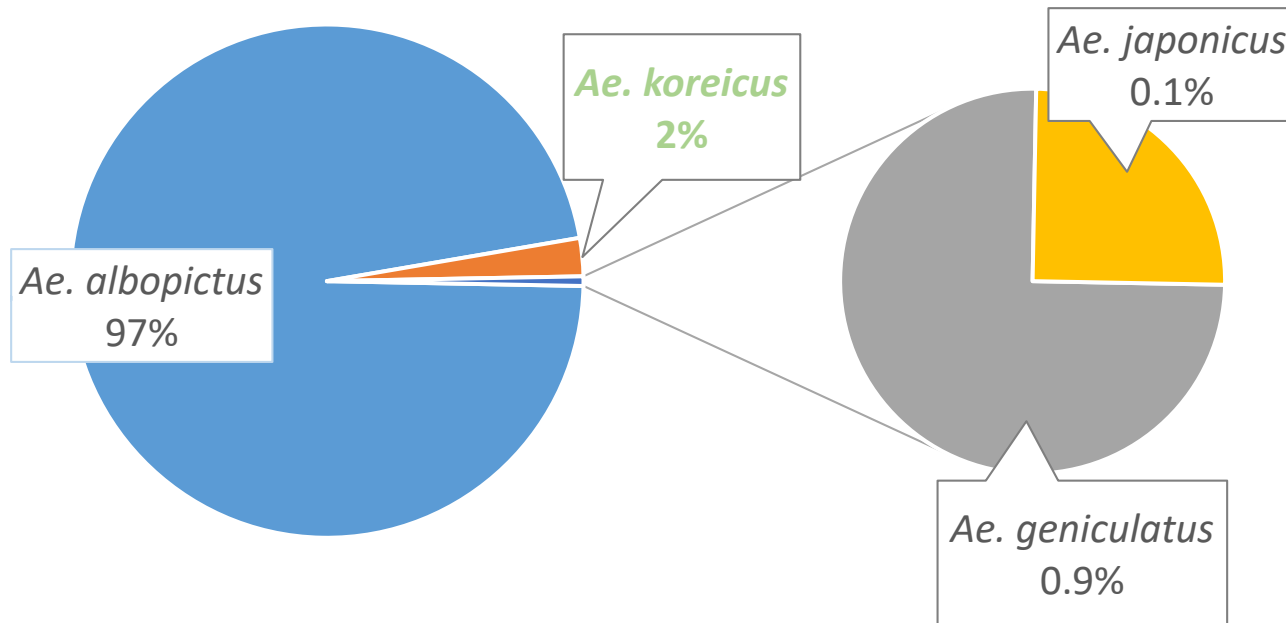
- OVI: Engaging 15 families for at-home ovitrap monitoring in 5 localities >600 m a.s.l. [Citizen Science]



Monitoring in numbers - Standard monitoring OVI (Trento Municipality <600 m asl)

N. OVI sites (2010-2025)		86
Tot. N. sticks		15,489
Tot. N. <i>Aedes</i> eggs (97% <i>Ae. albopictus</i>)		880,097

Weekly, from May to October



(unpublished)

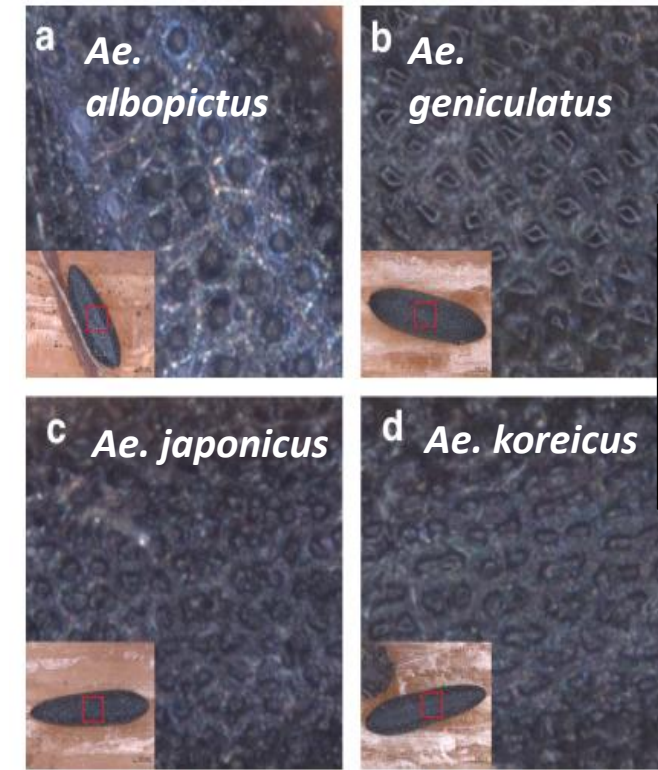
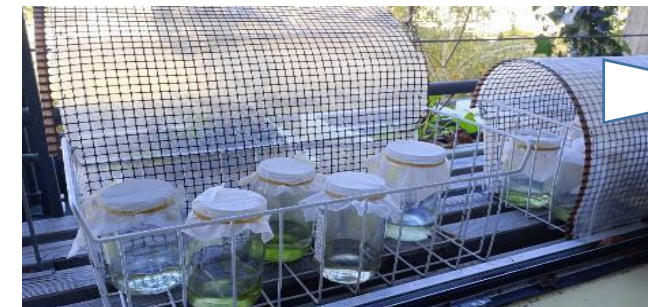
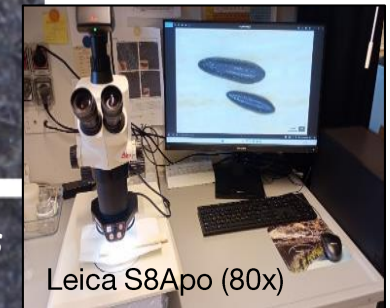
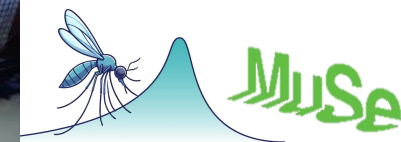


Fig 1. Patterns present on cuneate membrane. Egg of *Aedes albopictus* (a), *Aedes geniculatus* (b), *Aedes japonicus* (c), and *Aedes koreicus* (d).

PLOS ONE | <https://doi.org/10.1371/journal.pone.0293568> November 1, 2023



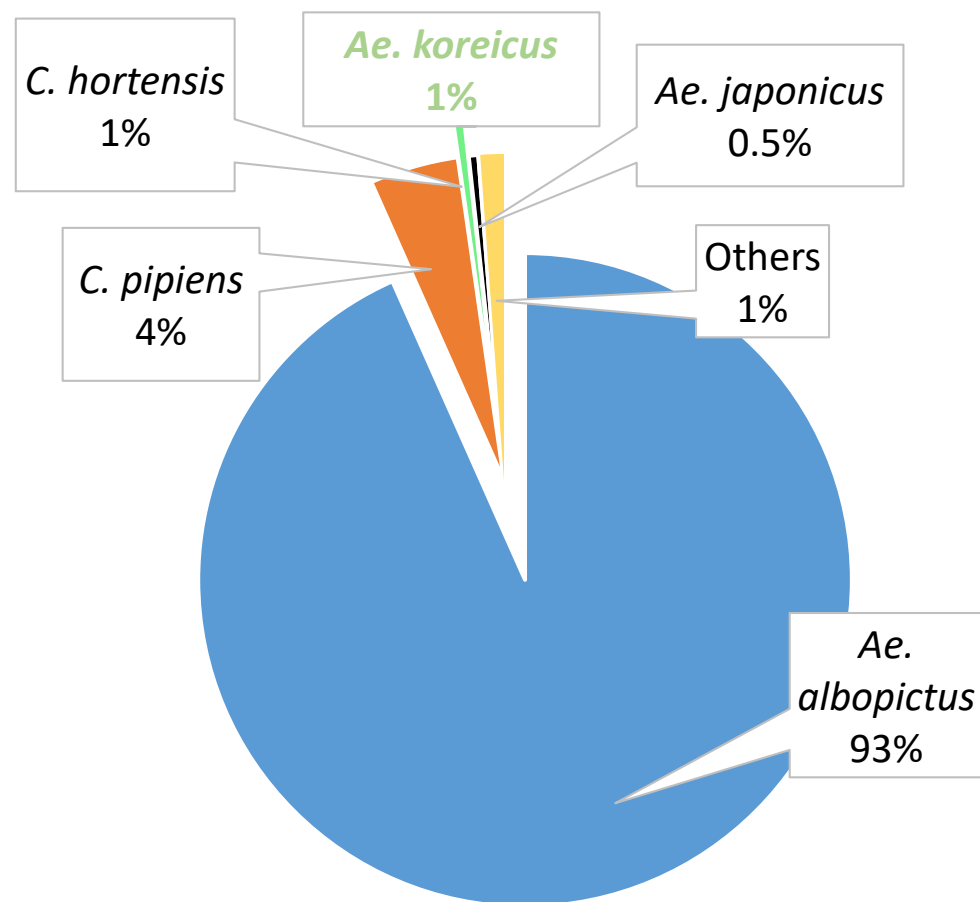
Egg rearing for species confirmation



Monitoring in numbers - BG-Sentinel + BG-GAT (Trento Municipality <600 m asl)



Biweekly monitoring, from May to October



Ae. albopictus

1990: In Italy - Genova;
1997: in Trentino



Ae. koreicus

2011: In Italy - Belluno;
2013: in Trentino



Ae. japonicus

2015: in Italy - Udine;
2021: in Trentino

MUSE's collection - Mosquito species list 2018-2025

<i>Aedes albopictus</i>	20,957
<i>Culex pipiens complex</i>	985
<i>Aedes koreicus</i>	122
<i>Culex hortensis</i>	105
<i>Aedes japonicus</i>	21
<i>Coquillettidia richiardii</i>	48
<i>Aedes geniculatus</i>	43
<i>Culex theileri</i>	42
<i>Culiseta longiareolata</i>	38
<i>Culex laticinctus</i>	29
<i>Culex impudicus</i>	23
<i>Culex brumpti</i>	15
<i>Culex perexiguus</i>	10
<i>Culex territans</i>	5
<i>Culex modestus</i>	4
<i>Anopheles spp.</i>	2
<i>Aedes communis</i>	1
<i>Culiseta annulata</i>	1
Total	22,451

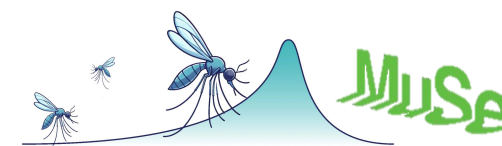
Others

ISSN: 1123-3117 (cartaceo) • 2384-8936 (online)

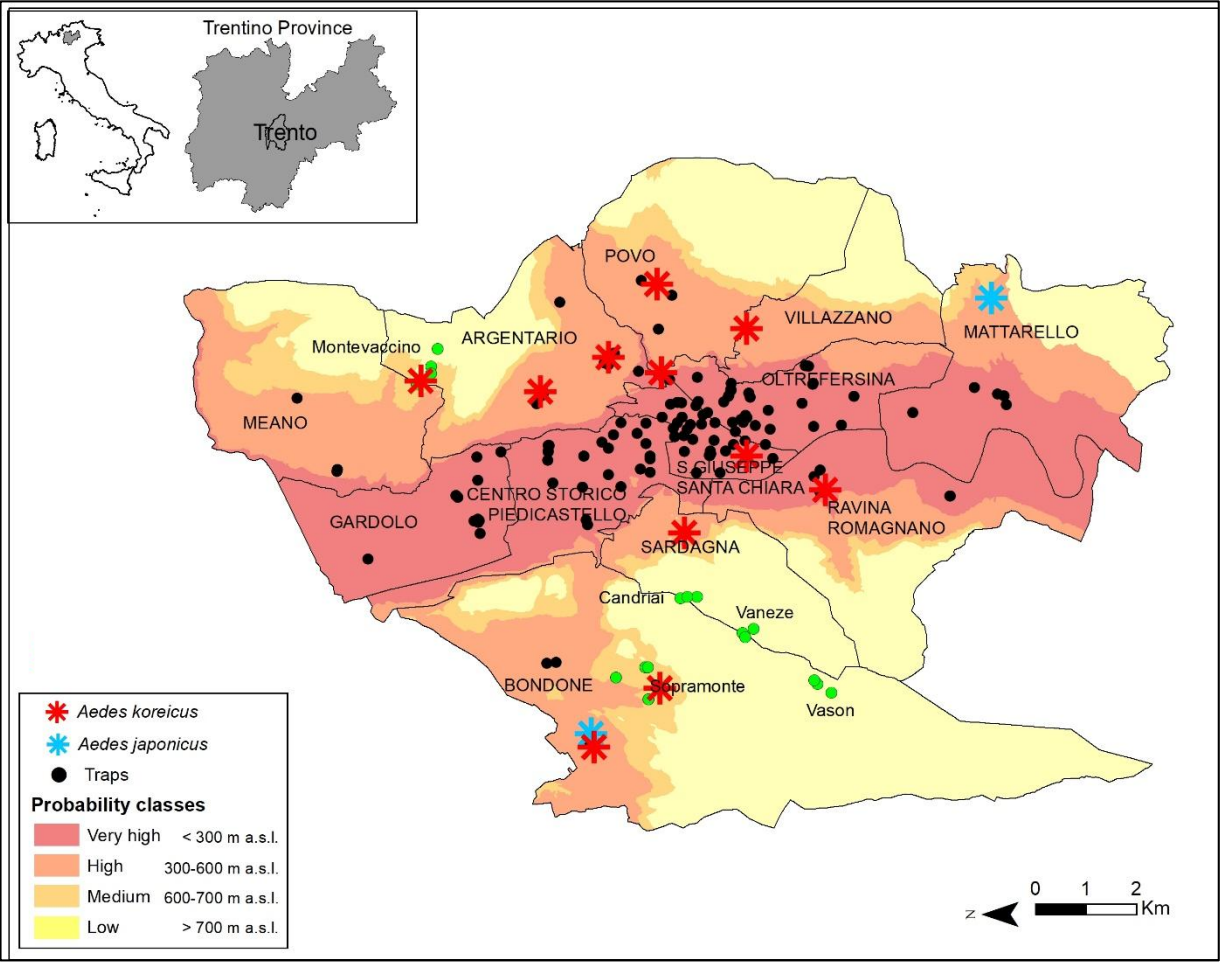
Rapporti ISTISAN
22/3

In ITALY: 65 mosquito
species (24 of *Aedes*)

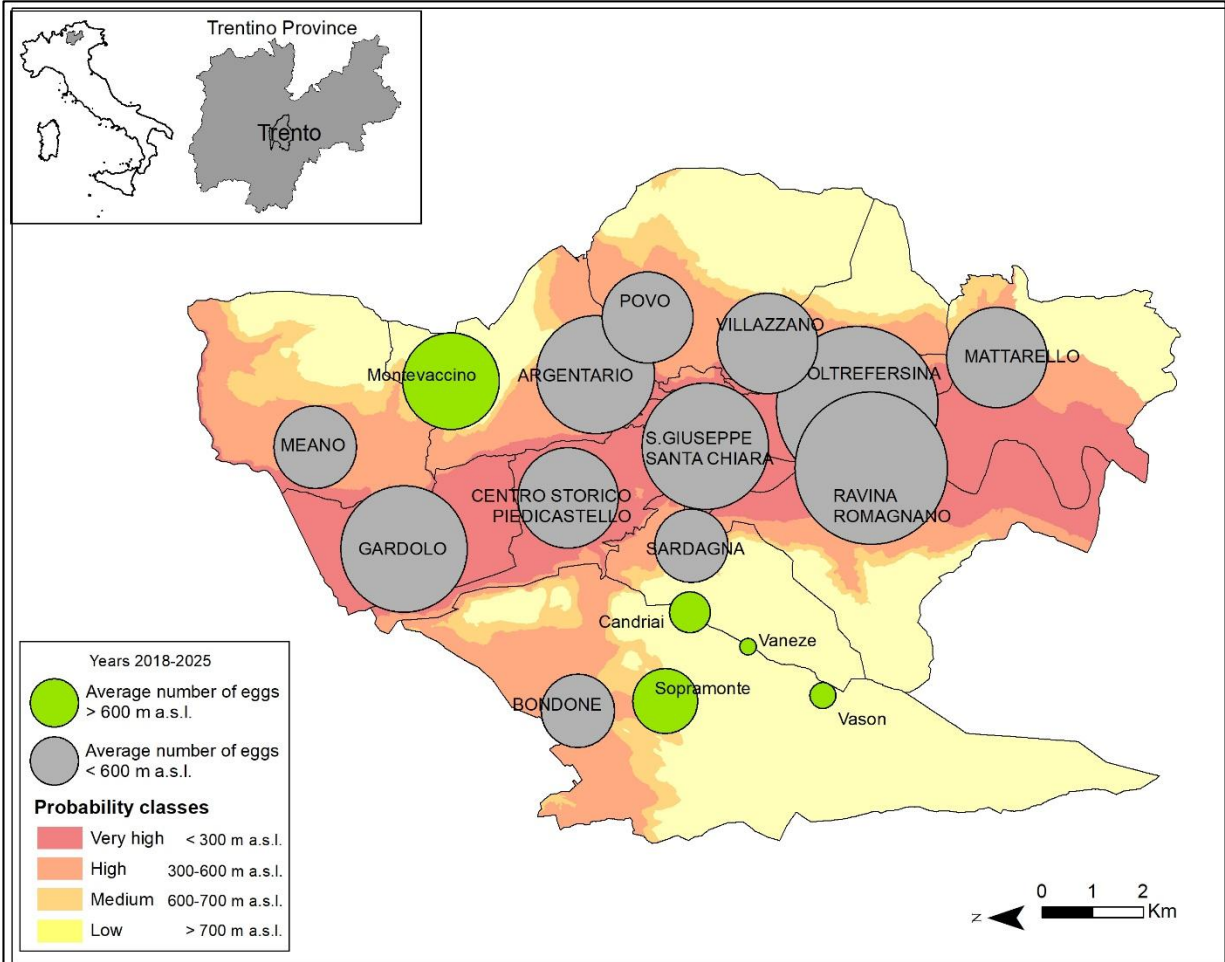
(unpublished)



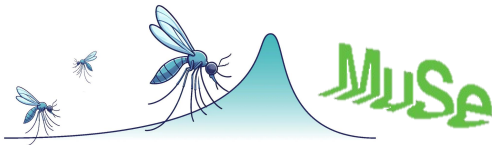
Alien Aedes species: Distribution and abundance



(unpublished)



(unpublished)



Monitoring data: use and enhancement

1- Weekly data: provide a description of the current situation & identification of potential outbreaks;

Real-time reporting: transmitted in real-time to the authorities and uploaded to the website

<https://www.muse.it/home/ricerca-e-collezioni/clima-ecologia/monitoraggio-della-zanzara-tigre-a-trento/>

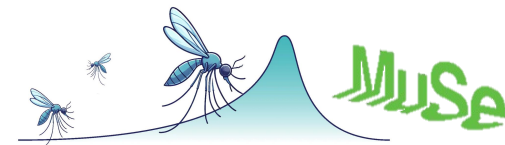
- to guide real-time municipal pest control by identifying high-risk areas before outbreaks occur
- to inform citizen in real time (data for the street where they live)
- to manage sanitary controls in emergency conditions



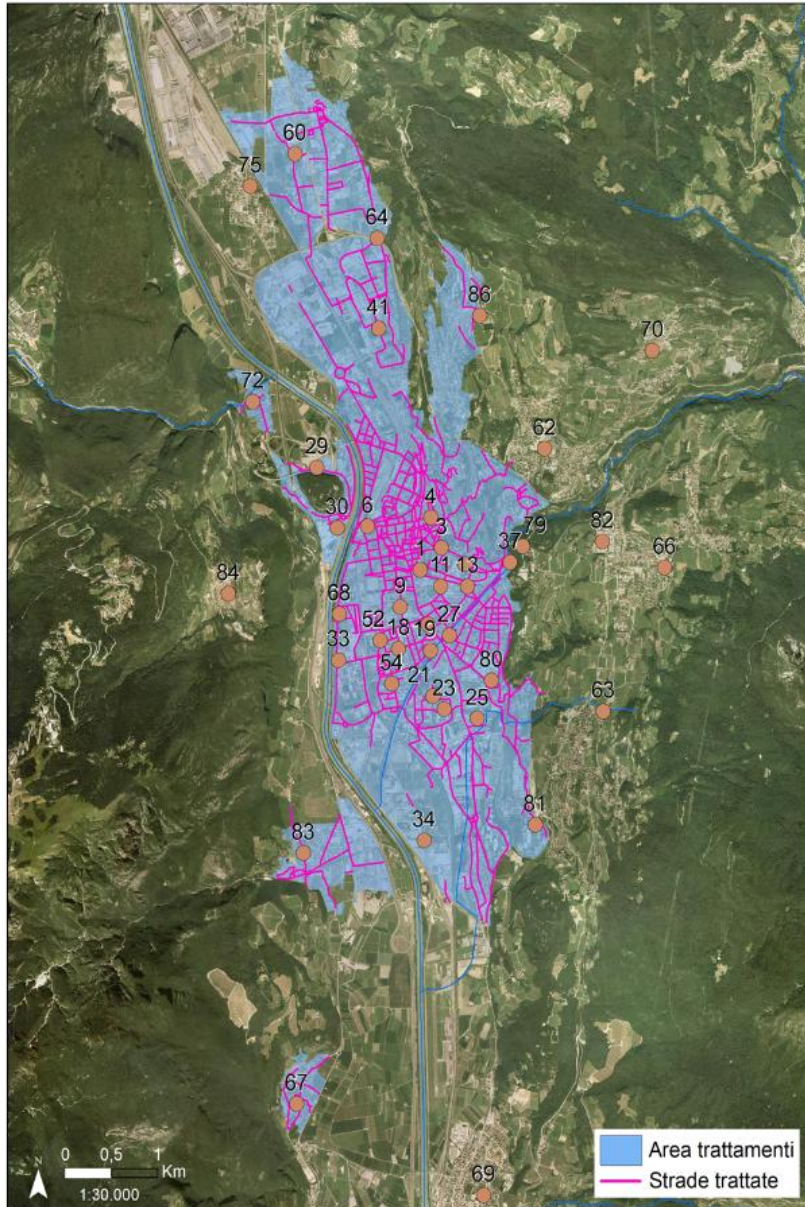
Imported cases of Dengue in Trento:

Site-specific snapshot of abundance data useful to plan **targeted Pest Control with adulticide** treatments

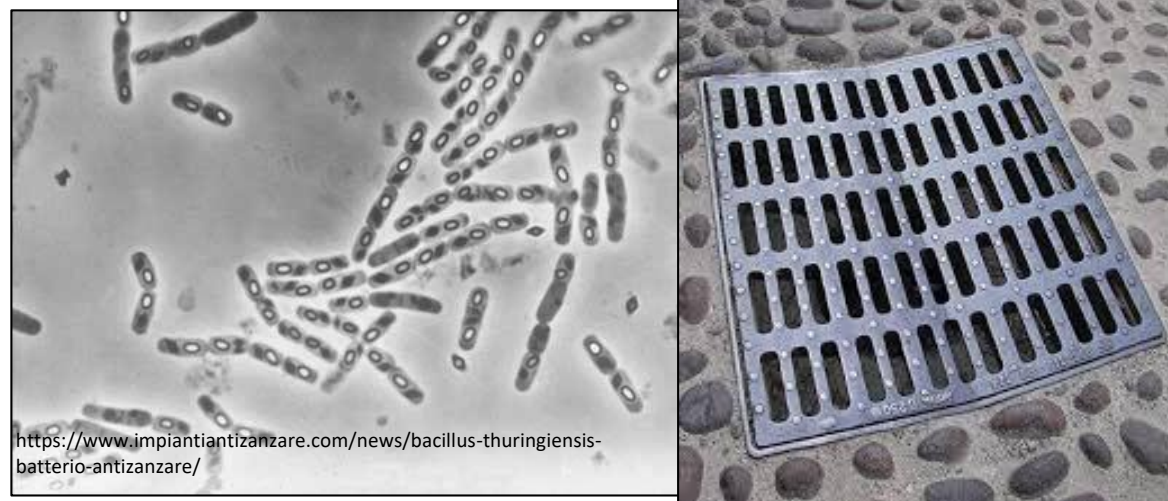
Specific extraordinary monitoring to evaluate the effectiveness of the treatments in the surrounding residential zone



Pest control: Municipality of Trento – Novareti spa (54,444 manholes, drains, grates in public areas)



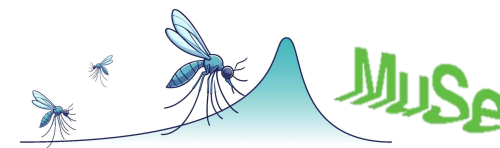
Since 2017, microbiological larvicide (VectoMax® FG) based on *Bacillus thuringiensis* var. *israelensis* and *Bacillus sphaericus* in granular form



- While studies have investigated Diflubenzuron resistance in Italian populations of *C. pipiens* and *Ae. albopictus*, the current data indicates that this resistance has not been found in the populations of Trento -

by avoiding routine adulticide spraying?

©Pichler et al. (2021). INSECTS <https://doi.org/10.3390/insects12010079>



Monitoring data: use and enhancement

2- Continuous time series, standardized, high-quality data on presence, absence, abundance **are essential** for **understanding the spatio-temporal dynamics** of an invasive alien species in relation to environmental/climatic factors

Generating **forecasts of egg abundance** even in **regions not covered by conventional monitoring** initiatives using a stacked **machine learning model** [Albania, France, Italy, and Switzerland]

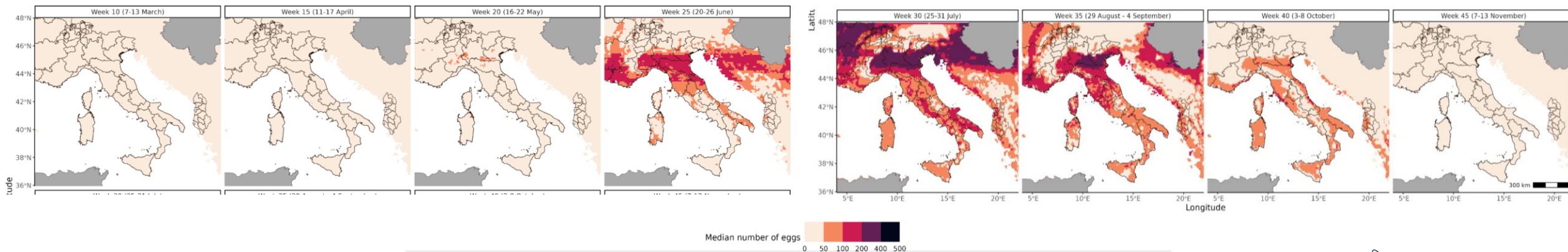
- to meet specific public health needs
- to plan pest control programs (*actionable prevention data*)

scientific **data**

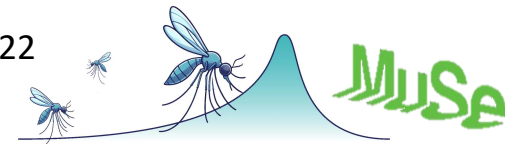
OPEN **VectAbundance: a spatio-temporal database of *Aedes* mosquitoes observations**
DATA DESCRIPTOR
Da Re et al. (2024), <https://doi.org/10.1038/s41597-024-03482-y>

scientific reports

OPEN **Modelling the seasonal dynamics of *Aedes albopictus* populations using a spatio-temporal stacked machine learning model**
Da Re et al. (2025), <https://doi.org/10.1038/s41598-025-87554-y>

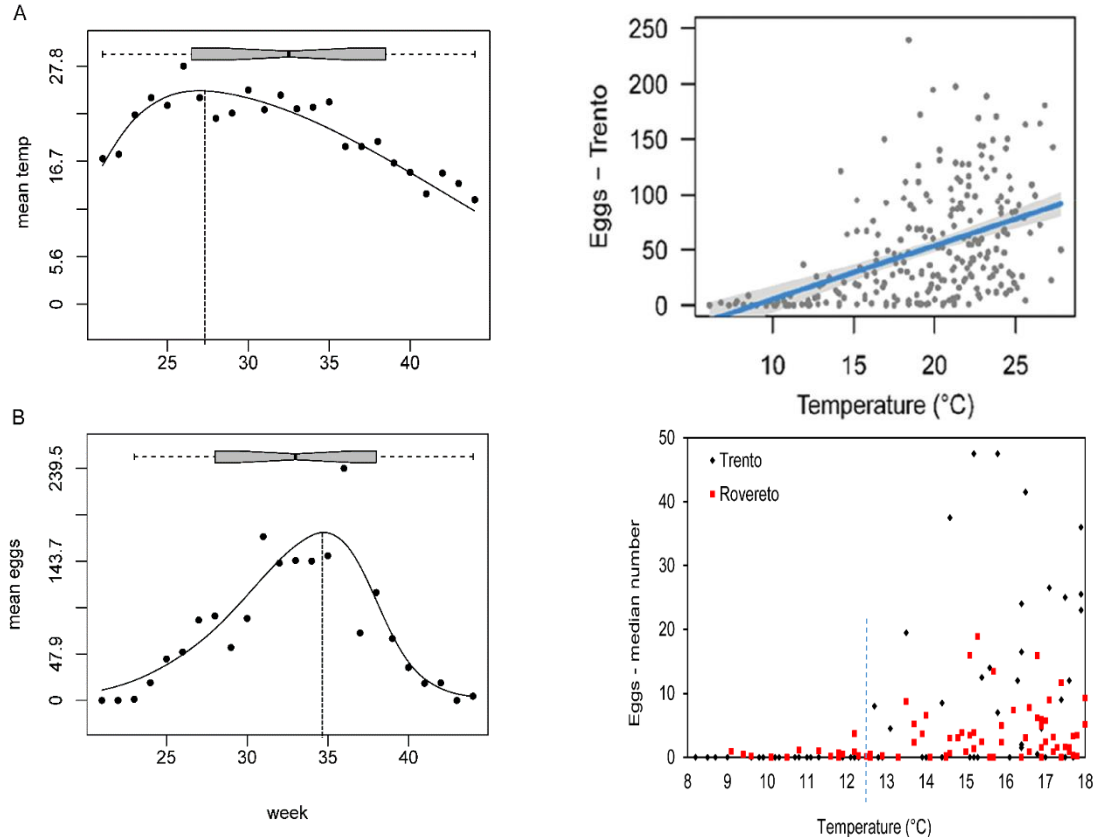


e.g. Median number of eggs predicted weekly by the regression model in the area of interest for the year 2022



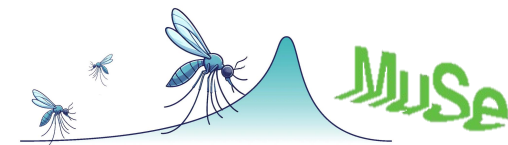
Monitoring data: use and enhancement

2- At a local scale: “Multi-year dynamics of the *Ae. albopictus* occurrence in Trento” - analysis temporal trend of **ovitrap data with climatic factors** in 2010-2020, inferring its **phenology**

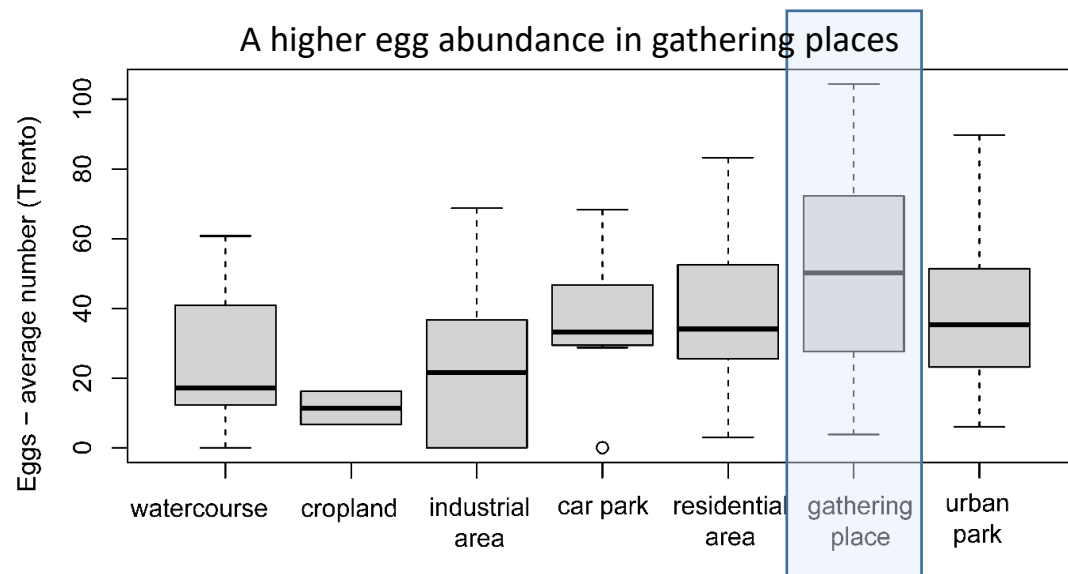


- **Linear and positive relationship between egg N and T**
- The **peak egg** occurs on av. **4.3-4.4 weeks after the peak of T**
- Oviposition begins at average weekly temperatures between 12.5°C and 13.5°C, establishing a **thermal activity threshold of ~ 13°C**
- The av. **N of eggs responds negatively to the precipitation** (i.e. high precipitation reduces short-term abundance of mosquitoes)

	Average air temperature (°C)			Tot rainfall (mm)		Average egg N/y
	Jan	May-Oct	Annual	May-Oct	Annual	May-Oct
2010-2020	1.8	19.5	12.8	602	1030	50,792
2021-2025	2.9	19.8	13.4	668	1055	64,278

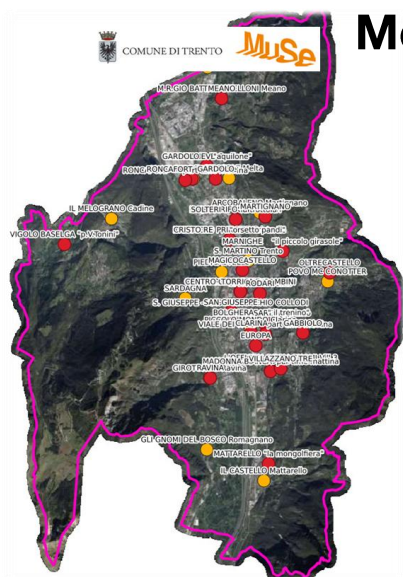


Community Engagement & Empowerment: kindergartens & nursery schools



In response to concerns and complaints from parents and teachers

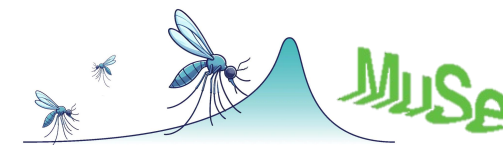
&



Monitoring: 46 kindergartens/nursery schools: ca 20,000 *Aedes* (99% *Ae. albopictus*) (2024-2025)



**& Training & Advisory Services
 for Schools and Families**
 (workshops, education,
 consulting)

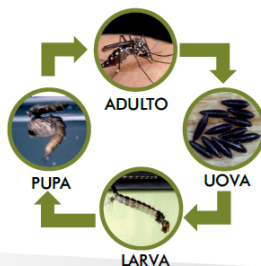


Community Engagement & Proximity: primary & secondary schools

Progetto **MOSQUITOS**

- Education projects for students and teachers
- Tutoring of students in **work-study programs**

Dissemination of «Best Practices»



IDENTIKIT

nome scientifico: *Aedes albopictus*
nome comune: zanzara tigre
origine: Asia; è in Italia dal 1990, in Trentino dal 1997
dimensioni: uovo 0,5 mm, adulto 4-10 mm
aspetto: nero con bande bianche sulle zampe e sull'addome e una striscia bianca sul capo e sul dorso
attività: da aprile a ottobre
riproduzione: le femmine depongono le uova in piccole raccolte d'acqua ferma (sulle pareti di bidoni, sottovasi, vasi, ecc.). In estate il ciclo vitale da uovo ad adulto si compie in circa una settimana
durata di vita: 3/4 settimane (la femmina in estate)
segni particolari: più aggressiva della zanzara comune, punge durante le ore diurne, anche attraverso i vestiti, spesso con punture ripetute e causa di ponfi dolorosi. Vola basso e non emette ronzio
salute: vettore di virus e febbri, per l'uomo e per gli animali domestici.



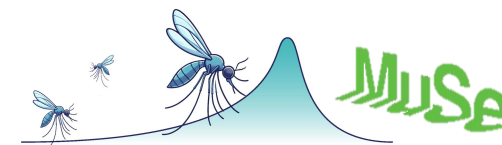
LE BUONE PRATICHE

- tenere pulite caditoie e griglie, coprirle con rete a maglia fine o inserire regolarmente un prodotto antilarvale
 - coprire con coperchio o rete a maglia fine tutti i tipi di contenitori nei quali può accumularsi acqua (bidoni, secchi, vasche, piscinette, ecc.)
 - svuotare settimanalmente sul terreno, o capovolgere se inutilizzati, annaffiatori, vasi, sottovasi
 - se non si può svuotare l'acqua di piccoli contenitori (come ad esempio vasi e sottovasi nei cimiteri) introdurre fili di rame da cambiare frequentemente oppure un prodotto antilarvale
 - favorire la lotta integrata con predatori naturali: ad esempio introdurre pesci ornamentali nelle fontane e nelle vasche dei giardini purché queste non abbiano scarico libero in torrenti o fiumi
 - non abbandonare né accumulare all'esterno oggetti che potrebbero diventare piccole raccolte d'acqua stagnante (bottiglie, lattine, bicchieri, tappi, copertoni ecc.)
- Alla zanzara basta un tappo per riprodursi!

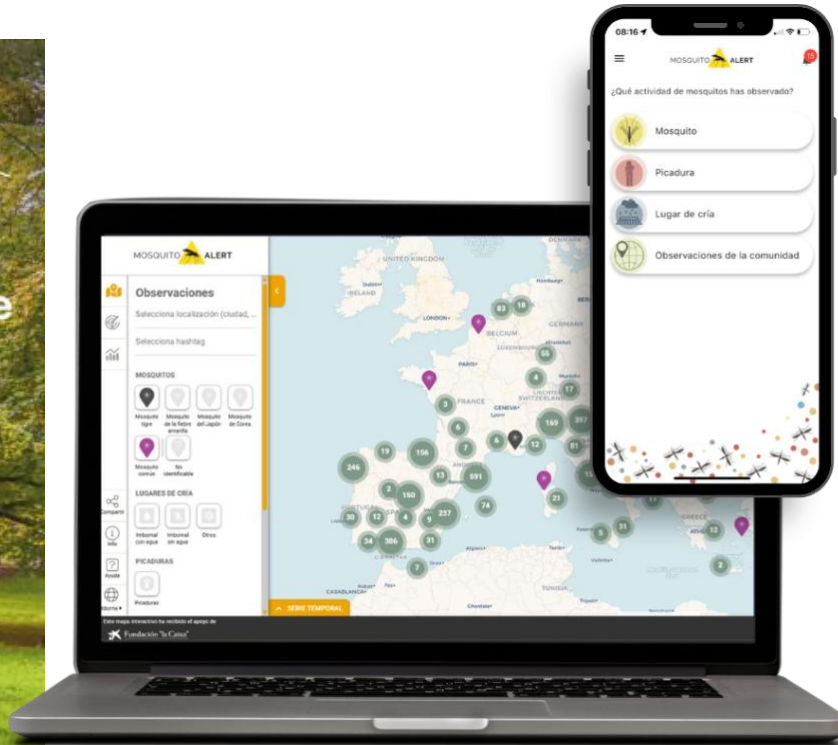
I principi attivi ad azione larvicida più efficaci sono di seguito riportati. Tutti i prodotti contenenti i seguenti principi attivi, reperibili presso le farmacie o altri rivenditori autorizzati, vanno usati e conservati seguendo le istruzioni del produttore, predisponendo tutte le adeguate misure di sicurezza.

Principi attivi utilizzo larvicidi:

- Diflubenzuron
- Piriproxyfen
- *Bacillus thuringiensis var. israelensis* (Bti)
- *Bacillus thuringiensis israelensis* + *Bacillus sphaericus* (Bti + Bs)
- Film monomolecolari



Citizen Science: the Mosquito Alert App & Home trap management



Mosquito Alert: to contain and control the spread of invasive mosquitoes through the use of a smartphone app – *in Trento: 14 breeding sites*

Adopted in 18 EU countries

Bringing citizens, scientists and managers of public health and the environment to fight against the tiger mosquito and other species, vectors of Zika, Dengue and Chikungunya

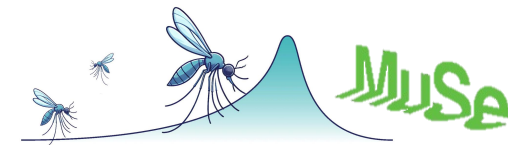
Journal of Entomological and Acarological Research 2024; volume 56:12185



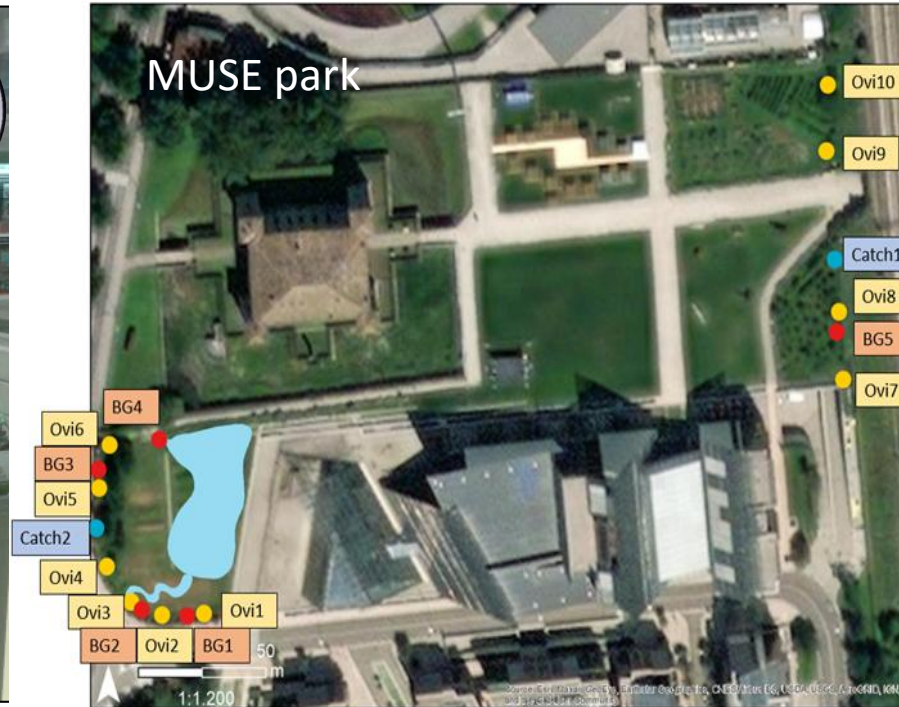
ENTOMOLOGY

Colonization by tiger mosquito (*Aedes albopictus* Skuse, 1894) of mountain areas over 600 m above sea level in the surroundings of Trento city, Northeast Italy

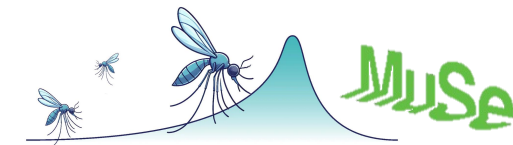
Giulia Battistin, Alessandra Franceschini, Francesca Paoli, Valeria Lencioni
MUSE – Museo delle Scienze, Research and Museum Collections Office, Trento, Italy



Dissemination with a participatory approach



- Open Museum Labs: rearing of *Ae. albopictus* - practical workshops on identification and making homemade traps
- Education activities at the MUSE and in the park MUSE
- Direct Citizen Support: An "entomo-point" provides specialized entomological consulting services for citizens and both public and private entities to address specific concerns
- Public events: e.g. Research Night, science cafés, conferences

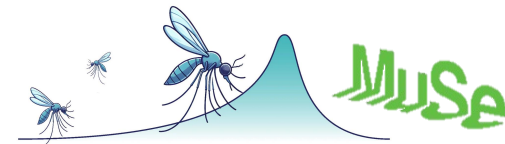


Conclusion and food for thought

- The spread of alien *Aedes* species is **not just a biological event but a public health priority – is it true also for *Ae. koreicus*?**
- In the fight against invasive vectors, **data collection is only the first step. The real challenge lies in how we bridge the gap between scientific findings and public action**



- ✓ **Optimizing Science Communication:** How and to what extent should we share technical data with the public?
- ✓ How can we reconcile complex scientific findings with **public perception ensuring transparency without triggering unnecessary fear?**
- ✓ What is **the best strategy for communicating** the presence of *Ae. koreicus* to the general public and to involve citizens in monitoring or prevention efforts?



Aknowledgements



- **Alessandra Franceschini**
- **Francesca Paoli**
- **Giulia Battistin**
- **Maria Vittoria Zucchelli**

MUSE



Paola Ricchi, Lino Nardelli, Lorenza Forti -
Sustainability and Ecological Transition
Service, Municipality of Trento (IT)



Pizzo Francesco -
Operational unit for hygiene and
public health, Health Agency of
Trentino Province, Trento (IT)

